

$\sqrt{p/\rho}$ has unit of m s^{-1}
 LHS = RHS so γ has no unit

B1
A0

- 2 (a) Using $v^2 = u^2 + 2as$

$$v = \sqrt{0 + 2(5.0)(4.0)}$$

$$= 6.32 \text{ m s}^{-1}$$

A1

- (b) Horizontal component $= 6.32 \cos 37.0^\circ$
 $= 5.05 \text{ m s}^{-1}$

A1

Initial vertical component of velocity $= 6.32 \sin 37.0^\circ = 3.80 \text{ m s}^{-1}$

$$\text{Using } v_y^2 = u_y^2 + 2gs_y$$

$$\text{Final vertical component of velocity} = \sqrt{3.80^2 + (2 \times 9.81 \times 15.0)}$$

$$= 17.6 \text{ m s}^{-1}$$

A1

- (c) Ball rolling down the roof: using $v = u + at$

$$t_1 = \frac{6.32 - 0}{5.00} = 1.264 \text{ s}$$

M1

Ball falling from edge of roof to ground: using $v_y = u_y + gt$

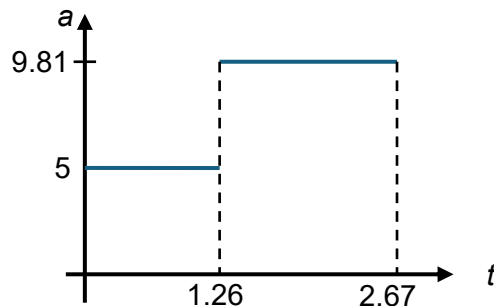
$$t_2 = \frac{17.57 - 3.80}{9.81} = 1.404 \text{ s}$$

M1

$$\text{Total time taken} = 1.264 + 1.404 = 2.668 = 2.67 \text{ s}$$

A1

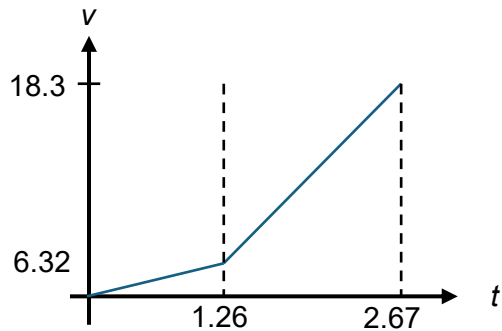
- (d) (i)



B1: Shape

B1: Correct labelled values (allow ecf)

(ii)



B1: Shape

B1: Correct labelled values (allow ecf)